

MedIntel

Use Case Diagram & UML Diagrams

AI-Assisted Symptom Triage, Health Records, and Medication Adherence

Course	Software Architecture — Semester 5
Authors	Rahul KT (2024BCS0224), Sathwik Teja E (2024BCD0040), Aryan Gupta (2024BCS0020), Sunil (2024BCS0340)
Document	MedIntel_UML.pdf
Source SRS	MedIntel_SRS.md, Version 1.0
Date	August 2026

Contents

1. Use Case Diagram
2. Use Case Descriptions
3. Class Diagram
4. Sequence Diagram — Submit Symptom Assessment
5. Activity Diagram — Symptom Analysis Workflow
6. State Diagram — Symptom Session Lifecycle

1. Use Case Diagram

MedIntel — Use Case Diagram

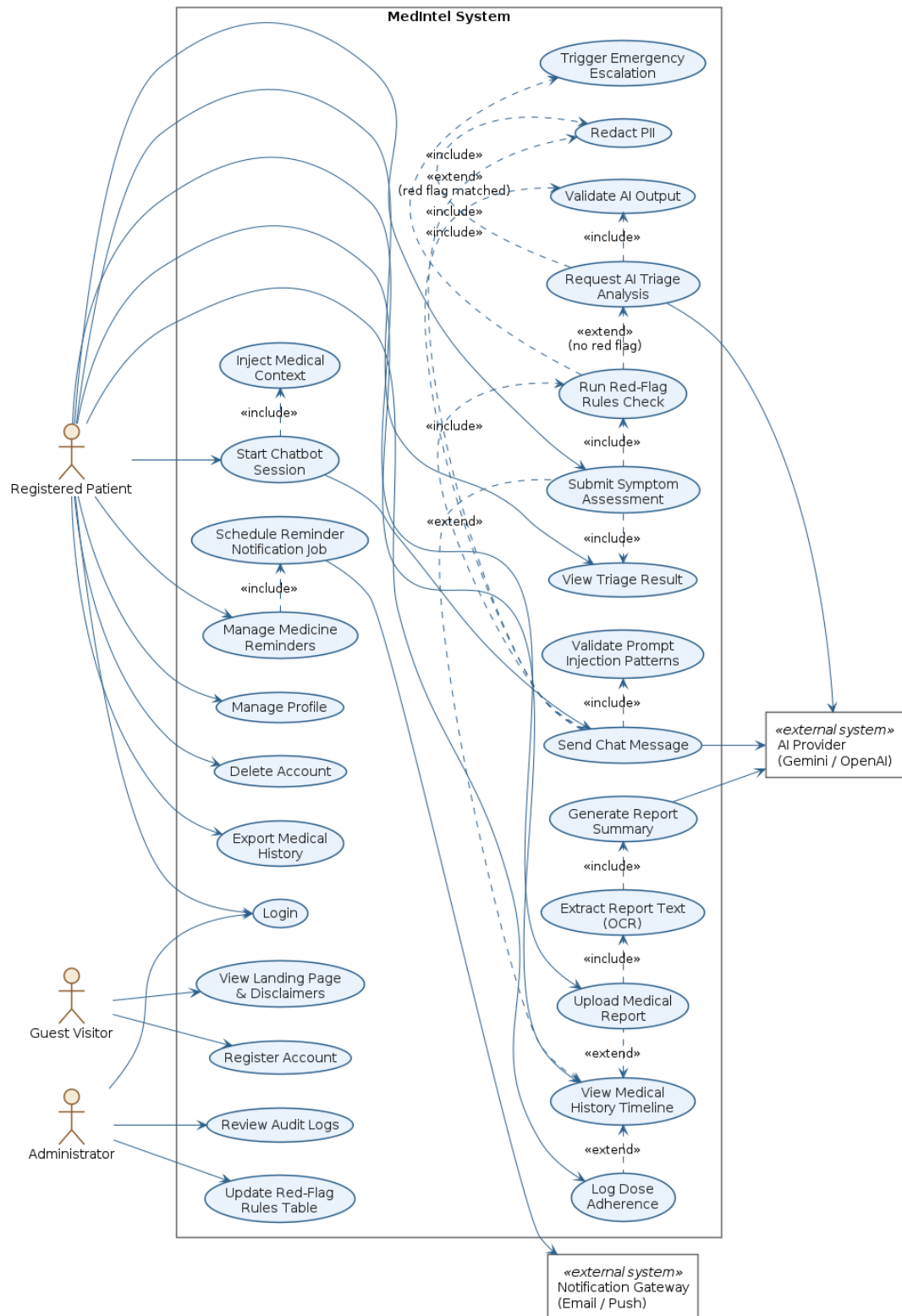


Figure 1: Actors — Registered Patient, Guest Visitor, and Administrator — interact with MedIntel's core use cases. The AI Provider and Notification Gateway are modelled as external systems. «include» edges denote mandatory sub-behaviour that always executes as part of the base use case; «extend» edges denote conditional behaviour that only executes under a stated condition (e.g. emergency escalation fires only when the red-flag check matches, while AI triage fires only when it does not).

2. Use Case Descriptions

View Landing Page & Disclaimers

Actor(s): Guest Visitor

Allows an unauthenticated visitor to view MedIntel's public landing page, including the mandatory non-diagnostic safety disclaimer. It requires no login and grants no access to personal data. The expected outcome is an informed visitor who understands the platform's scope before registering.

Register Account

Actor(s): Guest Visitor

Enables a new user to create a patient account using a unique email address and a password meeting strength criteria. Duplicate-email attempts return field-level validation errors (REQ-AUTH-1, REQ-AUTH-5). The outcome is a persisted user record with a securely hashed password, ready for authentication.

Login

Actor(s): Registered Patient, Administrator

Authenticates a user's credentials and issues a short-lived JWT access token together with a rotating refresh token (REQ-AUTH-3, ADR-05). It underlies every subsequent authorized action in the system. The outcome is a valid session token scoped to the user's role.

Manage Profile

Actor(s): Registered Patient

Lets a patient view and update profile attributes such as age, biological sex, allergies, and chronic conditions (REQ-AUTH-4). This data enriches later AI chatbot context. The outcome is an up-to-date profile used to personalise triage and chat interactions.

Delete Account

Actor(s): Registered Patient

Allows a patient to permanently remove their account. The system cascades the deletion across all associated symptom sessions, chat messages, reports, and reminders (REQ-HIST-4). The outcome is complete removal of the user's personal data from active storage.

Submit Symptom Assessment

Actor(s): Registered Patient

Captures structured symptom input — name, duration, severity, and onset pattern — through form components rather than free text (REQ-SYM-1). It always includes the Red-Flag Rules Check before any AI call. The outcome is a persisted symptom session with an associated triage result.

Run Red-Flag Rules Check

Actor(s): Registered Patient (system-invoked)

Evaluates a symptom submission or chat message against a deterministic Red-Flag Rules Table prior to any external AI invocation (REQ-SYM-2, ADR-02). It operates independently of AI provider availability. The outcome is a triage decision that routes the flow to either emergency escalation or AI analysis.

Trigger Emergency Escalation

Actor(s): Registered Patient (system-invoked)

Extends the Red-Flag Rules Check when a severe combination (e.g. chest pain with dyspnea) is matched. It halts AI execution entirely and returns immediate escalation instructions (REQ-SYM-3). The outcome is a fast, deterministic safety response and a corresponding audit log entry.

Request AI Triage Analysis

Actor(s): Registered Patient (system-invoked)

Extends the Red-Flag Rules Check when no red flag is matched, sending a PII-redacted payload to the AiProviderAdapter (REQ-SYM-4, REQ-SYM-7). It includes PII redaction and output validation as mandatory sub-steps. The outcome is a ranked set of condition possibilities with confidence and urgency indicators.

View Triage Result

Actor(s): Registered Patient

Presents the patient with ranked condition possibilities, an urgency level, a confidence score, and a mandatory physician-consult notice (REQ-SYM-5). It is the direct continuation of a submitted assessment. The outcome is a clearly non-diagnostic, actionable summary for the patient.

Start Chatbot Session

Actor(s): Registered Patient

Lets an authenticated patient begin a follow-up conversation linked to an active symptom session (REQ-CHAT-1). It includes injecting relevant profile and history context into the prompt (REQ-CHAT-2). The outcome is an initialised chat thread ready to receive messages.

Send Chat Message

Actor(s): Registered Patient

Passes each chat message through prompt-injection validation, PII redaction, red-flag parsing, and output validation before and after calling the external LLM (REQ-CHAT-3, REQ-CHAT-4). The outcome is a safe, context-aware reply persisted alongside token-usage metadata (REQ-CHAT-5).

Upload Medical Report

Actor(s): Registered Patient

Accepts PDF, JPG, or PNG report files up to a configured size limit (REQ-RPT-1). It includes asynchronous OCR extraction so the client is never blocked (REQ-RPT-2). The outcome is extracted text ready for AI summarisation, with the original file served only via signed URLs (REQ-RPT-5).

Manage Medicine Reminders

Actor(s): Registered Patient

Supports full CRUD operations on medicine reminders, including medication label, dosage notes, and recurring schedules (REQ-REM-1). It includes scheduling the underlying notification job. The outcome is an active or updated reminder whose jobs are immediately removed on deactivation (REQ-REM-5).

Log Dose Adherence

Actor(s): Registered Patient

Records whether a scheduled dose was taken or missed in response to a reminder notification (REQ-REM-3). It extends the medical history timeline with adherence events. The outcome is an updated adherence log entry contributing to the patient's unified timeline.

View Medical History Timeline

Actor(s): Registered Patient

Presents a unified, paginated, chronological timeline integrating symptom checks, reports, and adherence records (REQ-HIST-1), backed by compound indexes on `userId` and `createdAt` (REQ-HIST-3). The outcome is a single consolidated view of the patient's health activity.

Export Medical History

Actor(s): Registered Patient

Allows a patient to export their timeline data in a standardised format for clinical consultations (REQ-HIST-2). It reuses the same underlying history query as the timeline view. The outcome is a portable record explicitly noted as not constituting a legal health record (Business Rule 4).

Review Audit Logs

Actor(s): Administrator

Grants administrators read access to the append-only audit log covering logins, AI dispatches, and data deletions (Section 4.3, Business Rule 2). It is reserved exclusively for the Administrator role. The outcome is oversight visibility into security-relevant system events.

Update Red-Flag Rules Table

Actor(s): Administrator

Lets an administrator maintain the deterministic Red-Flag Rules Table that underlies emergency escalation (Section 2.2). Changes take effect for all subsequent red-flag evaluations. The outcome is an up-to-date safety ruleset pending required clinical sign-off (Appendix B.1).

3. Class Diagram

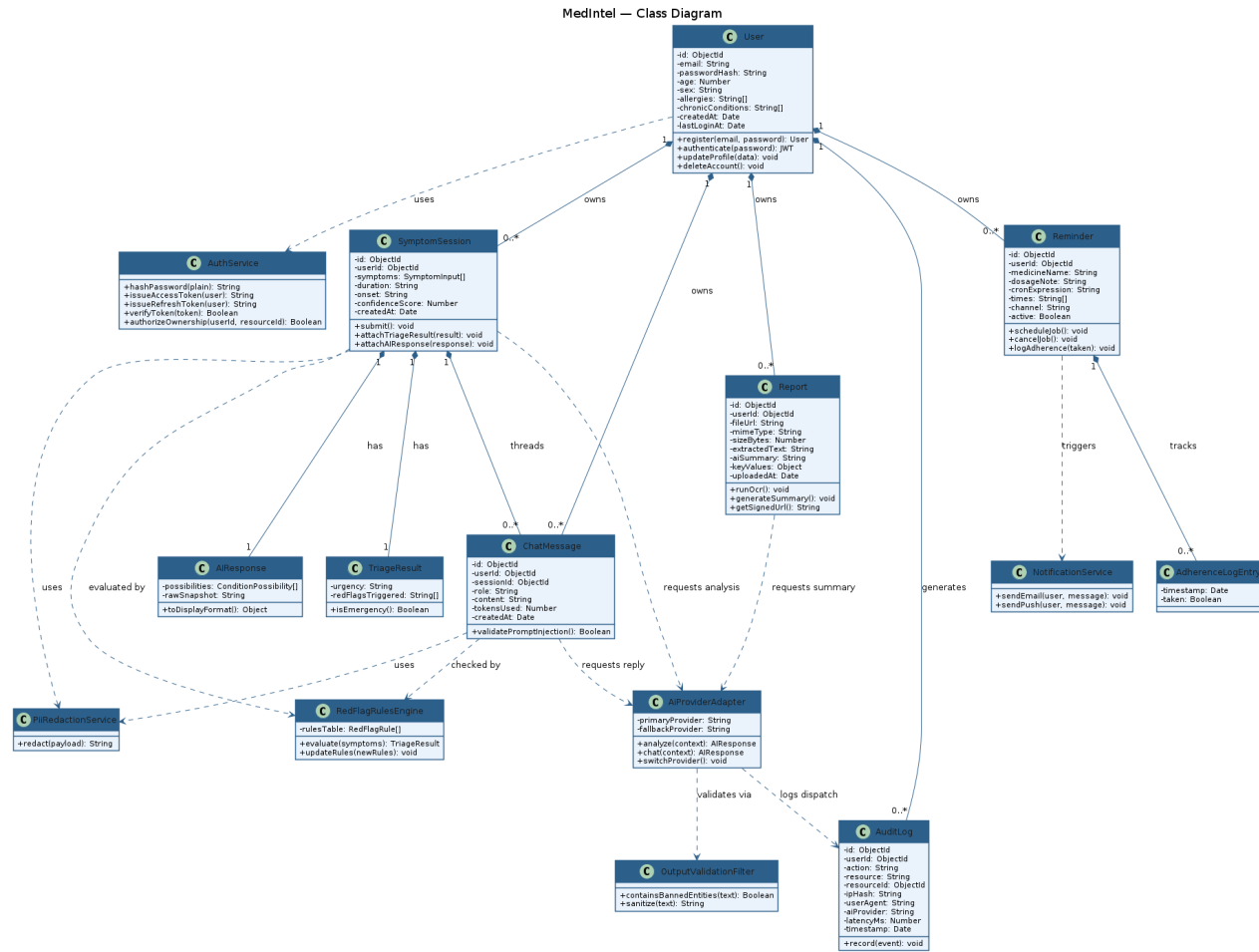


Figure 2: Core domain classes derived from the MongoDB collections in Section 5 of the SRS. User owns SymptomSession, ChatMessage, Report, Reminder, and AuditLog records. SymptomSession and ChatMessage both route through RedFlagRulesEngine, PiiRedactionService, and AiProviderAdapter, which in turn validates output via OutputValidationFilter — reflecting the safety-first pipeline mandated by ADR-02 and ADR-03.

4. Sequence Diagram — Submit Symptom Assessment

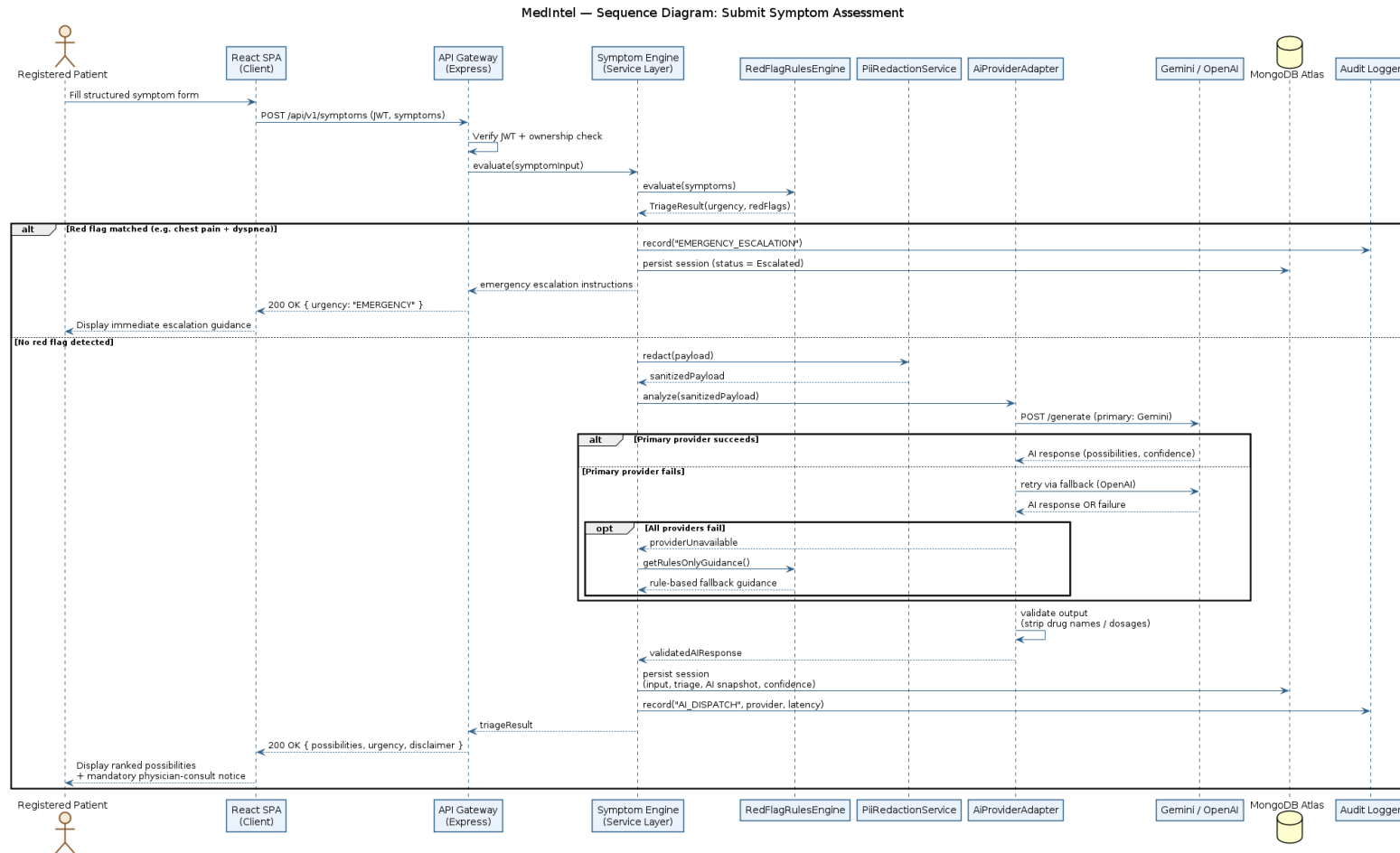


Figure 3: Chronological interaction for the Submit Symptom Assessment use case, the system's most safety-critical flow. The Red-Flag Rules Check always executes before any AI provider is contacted (ADR-02); a match halts AI execution and returns immediate escalation guidance, while a non-match proceeds through PII redaction, primary/fallback AI dispatch, output validation, and persistence.

5. Activity Diagram — Symptom Analysis Workflow

MedIntel — Activity Diagram: Symptom Analysis Workflow

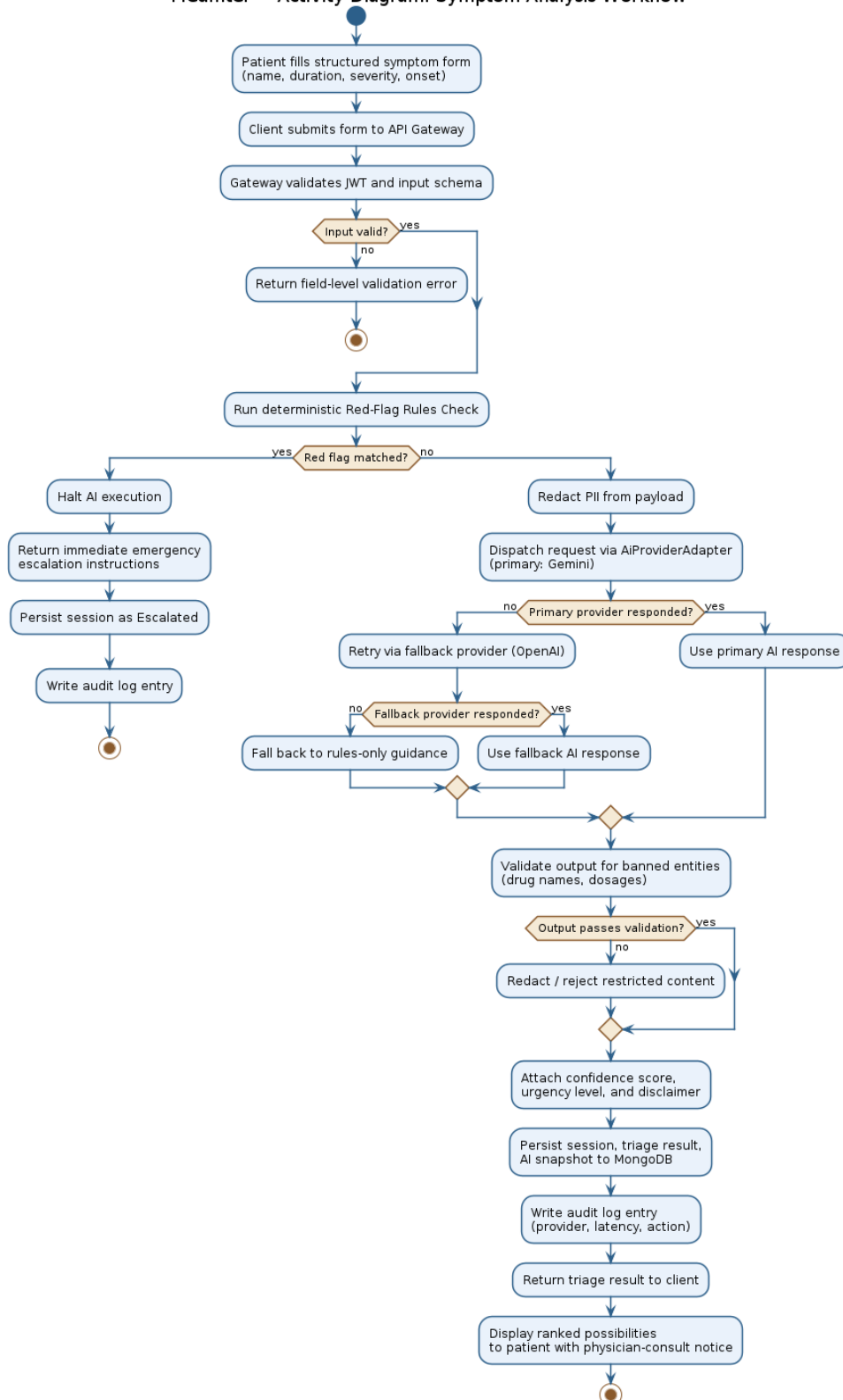


Figure 4: End-to-end workflow of the symptom analysis process, including the input-validation, red-flag, provider-failover, and output-validation decision points described in REQ-SYM-2 through REQ-SYM-7.

6. State Diagram — Symptom Session Lifecycle

MedIntel — State Diagram: Symptom Session Lifecycle

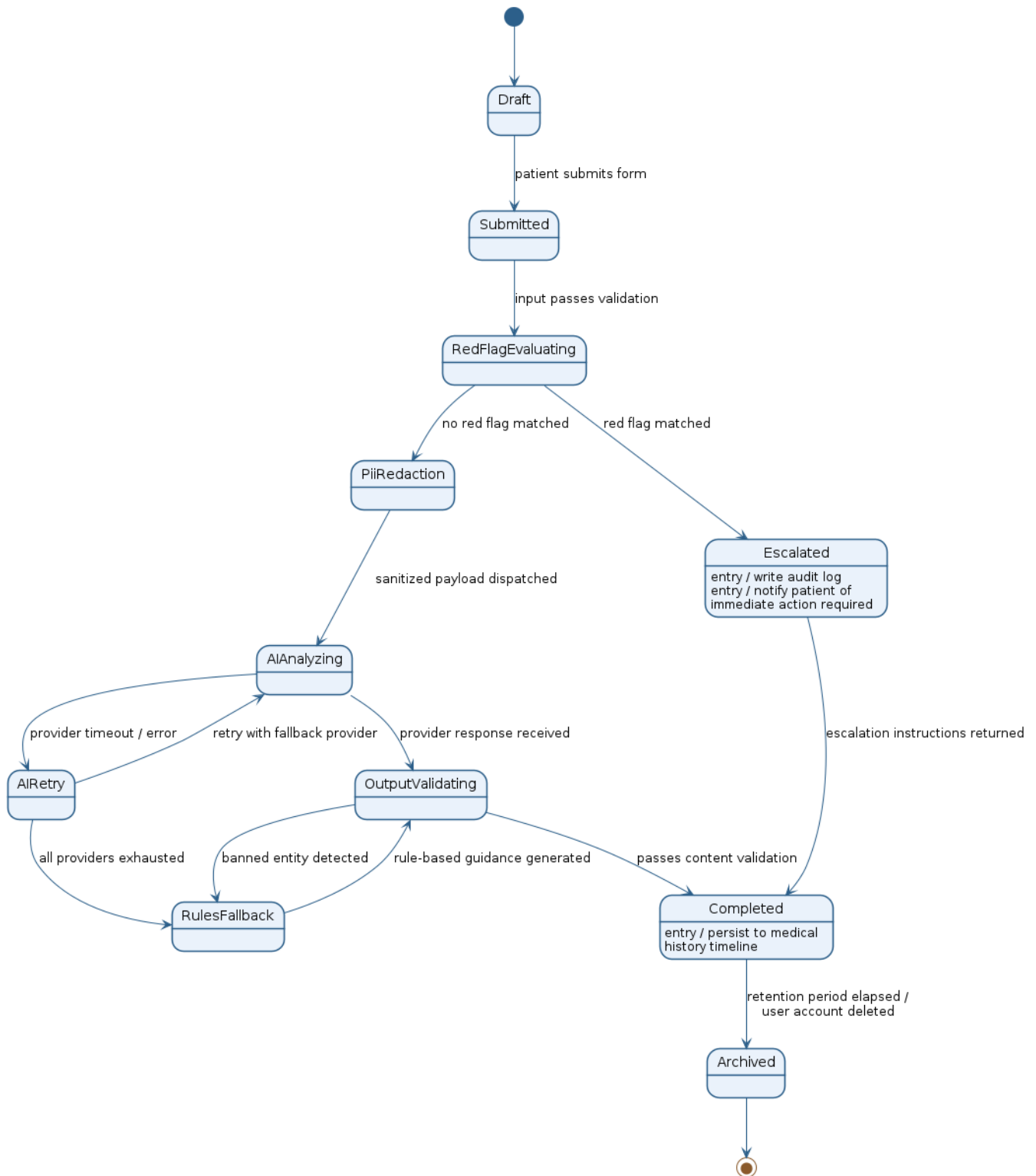


Figure 5: Lifecycle states of a SymptomSession object, from initial Draft through Red-Flag evaluation, either emergency Escalation or the PII-redaction / AI-analysis / retry / rules-fallback path, to final Completion and eventual Archival.